

# PHAR6302 — Weeks 1 & 2 MCQ Quiz

Twenty single-best-answer questions on cardiovascular disease risk assessment and hypertension. Answers with explanations begin on page 3. Content drawn from the 2023 Australian Guideline for Assessing and Managing CVD Risk, the Heart Foundation Guideline for the Diagnosis and Management of Hypertension in Adults (2016), and the Queensland Community Pharmacy CVD Risk Reduction Program protocols. Doses and specific drug combinations should be confirmed against Therapeutic Guidelines and AMH.

## Part A — Cardiovascular Risk Assessment

1. A 38-year-old non-Indigenous man with type 2 diabetes and no known cardiovascular disease asks about a heart check. Under the 2023 guideline, which applies?
  - a. He is too young for absolute risk assessment
  - b. He should be assessed, because diabetes lowers the starting age to 35
  - c. He should be managed as high risk without calculating
  - d. He should wait until age 45
2. Which finding means you should **not** use the AUS CVD Risk Calculator and should manage the patient as high risk immediately?
  - a. eGFR 52 mL/min/1.73m<sup>2</sup>
  - b. Urine ACR 18 mg/mmol in a man
  - c. Sustained eGFR 38 mL/min/1.73m<sup>2</sup>
  - d. Total cholesterol 6.2 mmol/L
3. Family history of premature cardiovascular disease is defined as coronary heart disease or stroke in a first-degree relative aged under:
  - a. 55 for women, 65 for men
  - b. 65 for women, 55 for men
  - c. 60 for both
  - d. 65 for both
4. A patient's calculated 5-year CVD risk is 4%. Which single factor would most appropriately move them **down** a category?
  - a. Coronary artery calcium score of 0
  - b. South Asian background
  - c. eGFR of 55 mL/min/1.73m<sup>2</sup>
  - d. Treated depression
5. A 55-year-old has a calculated 5-year risk of 7%. Which management approach matches the guideline?
  - a. Lifestyle only; no reassessment needed
  - b. Lifestyle, with pharmacotherapy considered depending on context; reassess in 2 years
  - c. Immediate BP- and lipid-lowering therapy; no formal reassessment
  - d. Lifestyle only; reassess in 5 years
6. Which combination of AUSDRISK score and 5-year type 2 diabetes risk is correct?
  - a. Score 7 → 1 in 30
  - b. Score 10 → 1 in 14
  - c. Score 13 → 1 in 14
  - d. Score 17 → 1 in 3
7. A patient has a coronary artery calcium score of 0 but a markedly elevated Lp(a). The most accurate interpretation is:
  - a. The zero score excludes coronary plaque and is reassuring
  - b. Lp(a) can be lowered with intensive lifestyle change, so the score will stay at 0
  - c. A zero score excludes only calcified plaque; high Lp(a) still confers high myocardial infarction risk
  - d. The calcium score should be repeated in 6 months

8. Which statement about the AUS CVD Risk Calculator is **incorrect**?

- a. Postcode is used to estimate socioeconomic disadvantage and can be adjusted
- b. It estimates risk over 5 years
- c. It has been validated for people with type 1 diabetes
- d. It has not been validated in people over 79 years

9. For Aboriginal and Torres Strait Islander people aged 18–29, the guideline recommends screening for individual risk factors. Which set is correct?

- a. Blood pressure and cholesterol only
- b. Smoking, blood pressure, blood glucose/HbA1c, eGFR, lipids, uACR and familial hypercholesterolaemia history
- c. Coronary artery calcium score and Lp(a)
- d. Absolute risk calculation using the standard calculator

10. The most effective way to communicate a 15% five-year risk to a patient with limited health literacy is:

- a. "Your risk is moderately high"
- b. "You have a 15% chance"
- c. "Out of 100 people like you, about 15 will have a heart attack or stroke in the next 5 years"
- d. "Your risk score is 15 on the calculator"

## Part B — Hypertension

11. A clinic reading of 164/98 mmHg is classified as:
- High-normal
  - Grade 1 (mild)
  - Grade 2 (moderate)
  - Grade 3 (severe)
12. Which home blood pressure average is diagnostic of hypertension?
- $\geq 130/80$
  - $\geq 135/85$
  - $\geq 140/90$
  - $\geq 120/70$
13. When measuring blood pressure, using a cuff that is too small for the patient's arm will:
- Underestimate blood pressure
  - Overestimate blood pressure
  - Have no consistent effect
  - Only affect the diastolic reading
14. An inter-arm systolic difference of 25 mmHg is found at a first visit. This should raise concern about:
- White-coat hypertension
  - Aortic dissection or subclavian stenosis
  - Atrial fibrillation
  - An incorrectly calibrated device
15. The mitral (apex) auscultation area is located at the:
- 2nd intercostal space, right sternal border
  - 2nd intercostal space, left sternal border
  - 4th intercostal space, lower left sternal border
  - 5th intercostal space, midclavicular line
16. A patient with newly diagnosed uncomplicated hypertension also has asthma and a history of gout. Which two agents are best avoided?
- ACE inhibitor and dihydropyridine CCB
  - Beta-blocker and thiazide diuretic
  - ARB and thiazide diuretic
  - Dihydropyridine CCB and beta-blocker
17. A patient develops bilateral ankle oedema 6 weeks after starting amlodipine. The most appropriate action is:
- Add a thiazide diuretic
  - Add frusemide
  - Reduce the dose or switch class; consider an ACEi/ARB which can offset the oedema
  - Continue unchanged, as this resolves spontaneously in all patients
18. Ten days after starting an ACE inhibitor, which result would be **expected** rather than concerning?
- A creatinine rise of about 20% with normal potassium
  - A creatinine rise of 45%
  - Potassium of 6.1 mmol/L
  - A fall in eGFR from 62 to 38
19. Under the Queensland Hypertension Clinical Protocol V2, which patient triggers **immediate referral**?
- A 72-year-old with BP 152/92 on one agent
  - A patient with eGFR 54 mL/min/1.73m<sup>2</sup>
  - A patient with LDL-C 4.2 mmol/L
  - A patient reviewed 4 weeks after a dose increase

**20.** A 62-year-old presents with BP 196/120, sudden slurred speech and facial droop. The correct interpretation and action is:

- a. Hypertensive urgency — oral therapy and GP review in 24–72 hours
- b. Grade 3 hypertension — arrange ambulatory monitoring
- c. Hypertensive emergency — call an ambulance for immediate hospital assessment
- d. White-coat effect — repeat after 5 minutes of rest

## Answers and Explanations

1. (b) Type 2 diabetes lowers the assessment starting age from 45 to 35. First Nations people are assessed from 30, with individual risk-factor screening from 18. Diabetes alone is not an automatic high-risk category, so the calculator is still used — although risk is roughly doubled, and more so with longer duration, poor control or microvascular complications.
2. (c) Automatic (clinically determined) high risk means sustained eGFR <45, urine ACR >25 mg/mmol in men or >35 in women, or confirmed familial hypercholesterolaemia. An eGFR of 52 and a uACR of 18 sit in the "impaired" band, where you calculate risk and then strongly consider reclassifying upward. Total cholesterol 6.2 is elevated but below the ≥7.5 threshold that should prompt consideration of FH.
3. (b) First-degree female relative under 65, or first-degree male relative under 55. The asymmetry catches people out — a father with a myocardial infarction at 60 does not meet the definition, but a mother with a stroke at 60 does.
4. (a) A CAC score of 0 and East Asian background are the two factors that reclassify risk downward. South Asian background, reduced eGFR and severe mental illness all reclassify upward. Note that downward reclassification is only appropriate when the calculated risk already sits close to a lower threshold, and a CAC of 0 does not reassure if Lp(a) is high.
5. (b) 7% falls in the intermediate band (5 to <10%). Lifestyle change is the cornerstone, medicines are considered depending on context, and reassessment is every 2 years (sooner if risk factors change). Low risk is reviewed every 5 years; high risk moves into ongoing clinical management without a formal reassessment interval.
6. (c) The AUSDRISK bands are: ≤5 → 1 in 100; 6–8 → 1 in 50; 9–11 → 1 in 30; 12–15 → 1 in 14; 16–19 → 1 in 7; ≥20 → 1 in 3. The score overestimates risk in people under 25 and underestimates it in Aboriginal and Torres Strait Islander people, who are considered high risk from age 18.
7. (c) A calcium score of zero means no calcium was detected in any plaque present — it does not mean there is no plaque. Soft, non-calcified plaque is vulnerable to rupture, and the Australian Atherosclerosis Society position statement notes that high Lp(a) carries high myocardial infarction risk at any calcium score including zero. Lp(a) is largely inherited and barely responds to lifestyle change.
8. (c) The calculator has *not* been validated for type 1 diabetes and may give an inaccurate estimate. The other three statements are correct: it is a 5-year estimate, postcode-based socioeconomic status is included and adjustable, and it is not validated above 79 years, where clinical judgement (frailty, comorbidity, life expectancy, preferences) takes over.
9. (b) The full screening set is smoking status, blood pressure, blood glucose or HbA1c, eGFR, serum lipids, urine ACR, and history of familial hypercholesterolaemia — ideally within a holistic health assessment with cultural safety, shared decision-making, informed consent and patient ownership of information. From age 30, CVD risk is assessed at an annual health check, opportunistically, or at least every 2 years.
10. (c) Natural frequencies are the most effective single format, and pairing them with a percentage or an icon array improves comprehension further. Vague verbal labels like "moderately high" mean different things to different people, and a bare number without a time frame is not interpretable.
11. (c) Grade 2 is 160–179 systolic or 100–109 diastolic. Grade 1 is 140–159 / 90–99 and Grade 3 is ≥180 / ≥110. Where systolic and diastolic fall in different grades, the higher grade applies — so 182/104 would be Grade 3 on the systolic alone.
12. (b) Home and daytime ambulatory thresholds are both ≥135/85. The 24-hour ambulatory average is ≥130/80 and night-time is ≥120/70. Clinic ≥140/90 is the trigger to *offer* out-of-clinic monitoring, not the out-of-clinic diagnostic threshold.
13. (b) A cuff that is too small overestimates blood pressure; one that is too large underestimates it. This is one of the commonest sources of measurement error, along with unsupported arms, crossed legs and insufficient rest.

**14. (b)** A difference above 20 mmHg is abnormal and can suggest aortic dissection or subclavian stenosis, warranting prompt medical assessment. A smaller difference above 5 mmHg is common and simply means you use the arm with the higher reading for all subsequent measurements.

**15. (d)** Aortic — 2nd ICS right sternal border; pulmonic — 2nd ICS left sternal border; Erb's point — 3rd ICS left; tricuspid — 4th/5th ICS lower left sternal border; mitral/apex — 5th ICS midclavicular line. Palpating the carotid while listening times S1 (with the upstroke) against S2 (after it).

**16. (b)** Beta-blockers are contraindicated in asthma and are not first-line for uncomplicated hypertension anyway. Thiazides cause dose-dependent hyperuricaemia and can precipitate gout. That still leaves three suitable first-line options: ACE inhibitor, ARB and dihydropyridine calcium channel blocker.

**17. (c)** Dihydropyridine oedema comes from precapillary arteriolar dilation, not fluid overload, so a diuretic will not correct it and adds electrolyte and volume risk for no benefit. Dose reduction, switching class, or adding an ACEi/ARB (which dilates the venous side and can offset the effect) are the appropriate responses.

**18. (a)** A modest creatinine rise of up to roughly 30% is expected as the efferent arteriole dilates and is acceptable if potassium stays normal. A larger rise, a marked eGFR fall or hyperkalaemia should prompt review — consider bilateral renal artery stenosis, hypovolaemia, or an NSAID interaction ("triple whammy" with a diuretic). Check renal function and electrolytes before starting and again 1–2 weeks after.

**19. (b)** An eGFR below 60 (or uACR  $\geq 3$  mg/mmol) is an immediate referral trigger. Others include BP  $\geq 180/110$ , HbA1c  $\geq 10\%$  or BGL  $\geq 20$ , total cholesterol  $\geq 7.5$  / LDL  $\geq 5.0$  / triglycerides  $\geq 6$ , suspected secondary hypertension, and failure to reach target after 4–6 weeks on two maximum-dose first-line agents. LDL-C 4.2 is below the referral threshold, and age 72 is within the 18–79 eligibility range.

**20. (c)** Severely elevated blood pressure with acute neurological deficit indicates a hypertensive emergency and acute stroke — this needs immediate hospital assessment and intravenous therapy, not oral medicines or outpatient follow-up. Urgency (option a) applies when BP exceeds 180/110 with symptoms such as severe headache but *without* acute organ dysfunction.

### Answer key

| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|----|----|----|----|----|----|----|----|----|----|
| b  | c  | b  | a  | b  | c  | c  | c  | b  | c  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| c  | b  | b  | b  | d  | b  | c  | a  | b  | c  |

Sources: 2023 Australian Guideline for Assessing and Managing CVD Risk (Heart Foundation); Heart Foundation Guideline for the Diagnosis and Management of Hypertension in Adults 2016; Queensland Community Pharmacy CVD Risk Reduction Program Hypertension Clinical Protocol V2; Ward et al., Heart Lung Circ 2023;32:287–296. These questions were written for revision and are not past exam material. Therapeutic Guidelines and AMH are subscription resources — confirm doses and combinations there.